ASSIGNMENT - 4

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SECTION: 'J'

COURSE: B.TECH CSE CORE

SUBJECT: DATA SCIENCE (21CSS303T)

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Q1)

Ans Data wrangling is the process of cleaning, transforming, and organizing raw data into a usable format for analysis

→ It's important because

- Raw data is often messy (missing values, duplicates, errors)
- Improves data quality → better analysis result
- Saves time during analysis.

→ Common steps

- Handling missing values
- Removing duplicates
- Converting data types
- Filtering irrelevant data

Q2)

Ans String manipulation means modifying and working text data using python

example

```
text = " Hellow world"

Print (text . lower() ) → Convert to lowercase

Print (text . upper() ) → Convert uppercase

Print (text. replace("world", "Python") ) → Replace word
```

Print (text.split()) → split string

Print (text.strip()) → Remove space

→ uses

- Data cleaning
- Text Processing
- NLP tasks

Q3

Ans Techniques used to process big data efficiently:

Key Practices

- Use efficient libraries (Pandas, Numpy)
- Use chunk processing (chunksize in pandas)
- Avoid loops → use vectorization
- Use parallel processing (Dask, multiprocessing)
- Optimize memory (correct data types)

Q4)

Ans Data transformation converts data into a suitable format for analysis or ML.

Common techniques

- Normalization → Scale values between 0 and 1
- Standardization → mean = 0, std = 1
- Encoding → convert categorical data (Label, One-hot)
- Aggregation → combining data (sum, mean)
- Feature scaling → improve model performance.